

Unified Military Analytics and Comparison Dashboard – Final Report

1. Introduction

Modern military analysis requires the integration of large datasets, meaningful indicators, and intuitive visualizations. Traditional static reports make it difficult to explore relationships between defense spending, manpower, and military assets across countries. To address this limitation, this project develops an interactive analytics platform for evaluating global military power in 2025.

The **Unified Military Analytics and Comparison Dashboard** provides a comprehensive system for analyzing and comparing military strength across more than 140 countries. The system integrates data scraping, data preprocessing, KPI engineering, and interactive dashboard development into a unified analytical platform.

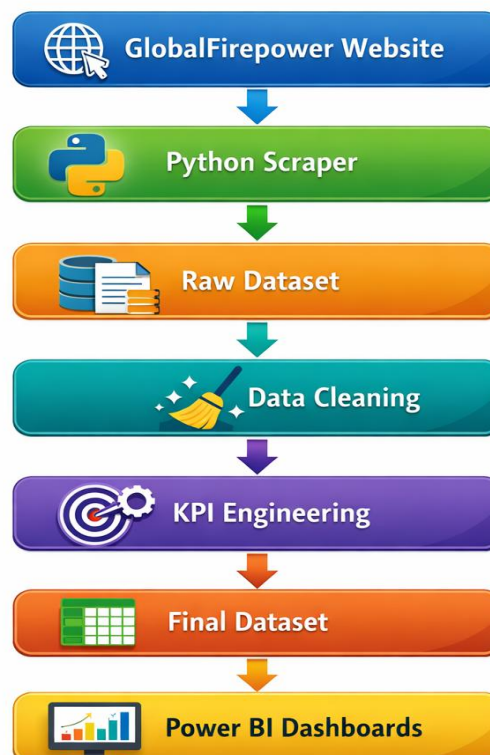
The project uses Python-based data pipelines to collect and process military data from [GlobalFirepower.com](https://www.globalfirepower.com). The processed dataset is then used to build an interactive dashboard using Power BI.

The platform enables users to:

- Analyze global military trends
- Examine detailed country profiles
- Compare military powers
- Simulate coalitions
- Evaluate defense efficiency

The final system combines more than 50 defense and economic indicators into an integrated analytical environment.

System Architecture Diagram:



2. Module 1 – Scraping Setup and Execution

2.1 Objective

The objective of Module 1 was to build a reliable scraping system capable of collecting military data for more than 140 countries from GlobalFirepower.com.

The scraping system was implemented in Python using:

- requests
- BeautifulSoup
- pandas

The script automatically collected country-level military metrics from a predefined list of URLs.

2.2 URL-Based Scraping Approach

A text file named:

links_for_military_data.txt was used as the source of all URLs.

Each link corresponded to a specific military metric such as:

- Total aircraft
- Tanks
- Naval fleet
- Defense budget
- Manpower

This ensured consistent and reproducible data collection.

2.3 Scraping Logic

The scraping script performed the following steps:

- Load URLs from text file
- Send HTTP request to each page
- Parse HTML structure
- Extract country names and values
- Store results in structured format

The script was implemented in:

scrape_military_metrics.py

```
def get_soup(url):
    response = requests.get(
        url,
        headers={"User-Agent": "Mozilla/5.0"},
        timeout=20
    )
    response.raise_for_status()
    return BeautifulSoup(response.text, "html.parser")

get_soup(BASE_URL)

<!DOCTYPE html>

<html lang="en-US">
<head>
<meta charset="utf-8"/>
<title>2025 Military Strength Ranking</title>
<link href="https://www.globalfirepower.com/countries-listing.php" rel="canonical"/>
<link href="https://www.globalfirepower.com/imgs/design/logo.png" rel="image_src"/>
<style type="text/css">@font-face {font-family:IBM Plex Sans;font-style:normal;font-weight:100;src:url(/cf-fonts/s/ibm-plex-sans/5.0.18/latin-ext/100/normal.woff2);unicode-range:U+0100-02AF,U+0304,U+0308,U+0329,U+1E00-1E9F,U+1EF2-1EFF,U+2020,U+20A0-20AB,U+20AD-20CF,U+2113,U+2C60-2C7F,U+A720-A7FF;font-display:swap;}@font-face {font-family:IBM Plex Sans;font-style:normal;font-weight:100;src:url(/cf-fonts/s/ibm-plex-sans/5.0.18/cyrillic-ext/100/normal.woff2);unicode-range:U+0460-052F,U+1C80-1C88,U+20B4,U+2DE0-2DFF,U+A640-A69F,U+FE2E-FE2F;font-display:swap;}@font-face {font-family:IBM Plex Sans;font-style:normal;font-weight:100;src:url(/cf-fonts/s/ibm-plex-sans/5.0.18/vietnamese/100/normal.woff2);unicode-range:U+0102-0103,U+0110-0111,U+0128-0129,U+0168-0169,U+01A0-01A1,U+01AF-01B0,U+0300-0301,U+0303-0304,U+0308-0309,U+0323,U+0329,U+1EA0-1EF9,U+20AB;font-display:swap;}@font-face {font-family:IBM Plex Sans;font-style:normal;font-weight:100;src:url(/cf-fonts/s/ibm-plex-sans/5.0.18/latin/100/normal.woff2);unicode-range:U+0000-00FF,U+0131,U+0152-0153,U+02BB-02BC,U+02C6,U+02DA,U+02DC,U+0304,U+0308,U+0329,U+2000-206F,U+2074,U+20AC,U+2122,U+2191,U+2193,U+2212,U+2215,U+FEFF,U+FFFD;font-display:swap;}@font-face {font-family:IBM Plex Sans;font-style:normal;font-weight:100;src:url(/cf-fonts/s/ibm-plex-sans/5.0.18/greek/100/normal.woff2);unicode-range:U+0370-037F;font-display:swap;}@font-face {font-family:IBM Plex Sans;font-style:normal;font-weight:200;src:url(/cf-fonts/s/ibm-plex-sans/5.0.18/latin-ext/200/normal.woff2);unicode-range:U+0100-02AF,U+0304,U+0308,U+0329,U+1E00-1E9F,U+1EF2-1EFF,U+2020,U+20A0-20AB,U+20AD-20CF,U+2113,U+2C60-2C7F,U+A720-A7FF;font-display:swap;}@font-face {font-family:IBM Plex
```

2.4 Raw Dataset Generation

The extracted data was stored in:

military_raw_data.csv

The dataset contained:

- 145 countries
- 55 metrics

2.5 Challenges Encountered

Several challenges were addressed:

- Dynamic HTML structure
- Some pages did not contain tables and required class-based extraction.
- Missing values
- Certain countries had incomplete data.
- Formatting issues

Numbers contained:

- commas
- symbols
- text labels

These issues were addressed in Module 2.

3. Module 2 – Data Cleaning and Structuring

3.1 Objective

The objective of Module 2 was to convert raw scraped data into a structured and analysis-ready dataset.

3.2 Data Cleaning Steps

Cleaning included:

- Text Cleaning

Removed:

- commas
- percentage symbols
- plus signs
- text annotations

Example:

"1,250,000" → 1250000

Data Type Conversion

- Most columns were initially stored as text.
- They were converted to numeric format.

Example:

Defense Budget → float

Personnel → integer

Column Standardization

Column names were standardized:

Example:

Total Military Aircraft → total_military_aircraft

This improved readability.

Missing Value Handling

Missing values were replaced using:

- zeros for equipment
- averages for ratios

After cleaning:

Missing values < 2%

3.3 Clean Dataset

Final cleaned dataset:

- military_cleaned.csv
- Prepared for KPI engineering.

4. Module 3 – KPI Feature Engineering

4.1 Objective

The objective of Module 3 was to derive meaningful indicators from raw metrics.

Raw numbers alone cannot accurately represent military strength. Derived indicators provide deeper insight.

4.2 KPIs Created

1. Assets per Capita

Total Assets / Population

Measures militarization level.

2. Budget-to-GDP Ratio

Defense Budget / Purchasing Power Parity

Measures economic commitment.

3. Budget per Active Personnel

Defense Budget / Active Personnel

Measures spending efficiency.

4. Air–Land–Naval Balance Ratio

Uses:

- aircraft
- tanks
- naval fleet
- Measures military structure.

5. Coalition Share Metrics

Coalition Assets / Global Assets

Measures coalition strength.

4.3 Metadata Enrichment

Additional metadata fields were added:

- Region
- Continent
- NATO membership

4.4 Final Dataset

Exported as:

- military_final.xlsx

Generated using:

- generate_kpis.py

5. Module 4 – Dashboard Planning and Prototyping

5.1 Objective

The objective of Module 4 was to design the dashboard structure before implementation.

5.2 Dashboard Structure

Four dashboards were planned:

- Quick Stats
- Nation Overview
- Compare Powers
- Coalition Builder

Each dashboard serves a different analytical purpose.

Initial Dashboard Layout:



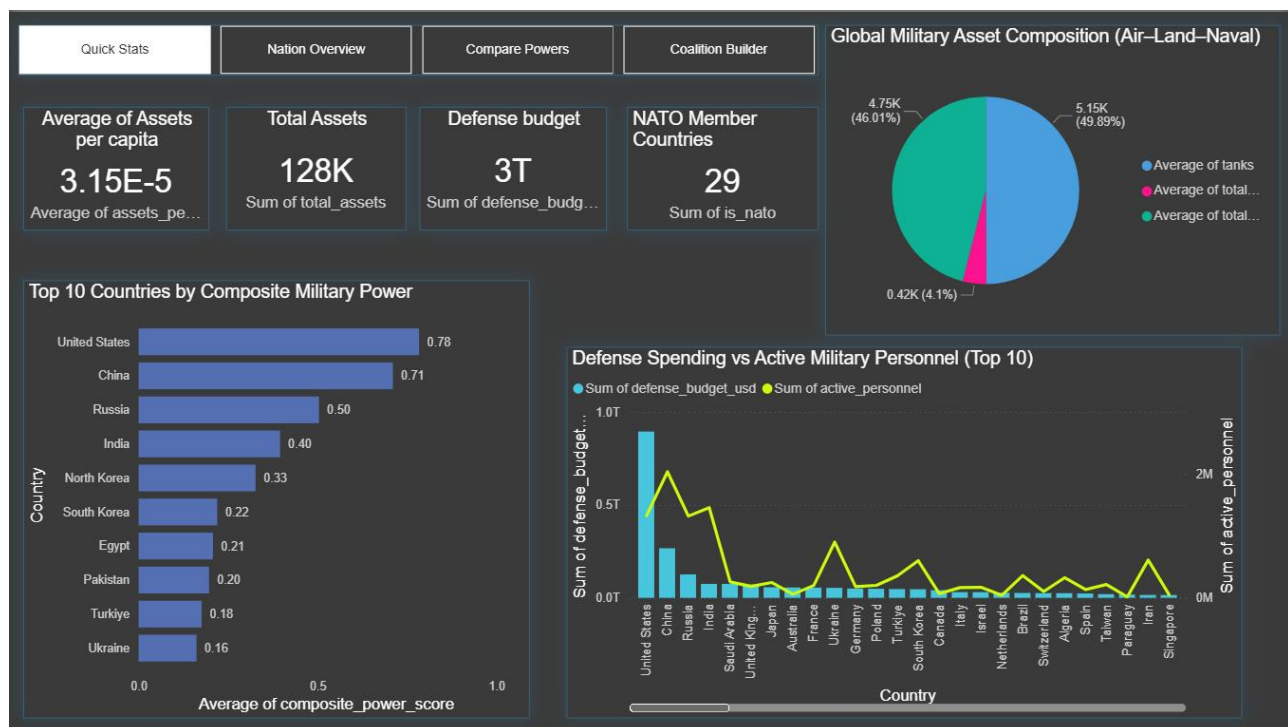
6. Module 5 – Quick Stats and Nation Overview

6.1 Quick Stats Dashboard

Provides global overview.

Includes:

- Global defense budget
- Total assets
- NATO count
- Top 10 military powers
- Spending vs personnel
- Asset composition



6.2 Nation Overview Dashboard

Provides country profile.

Includes:

- Country selection
- Personnel
- Budget
- Assets
- NATO status
- Asset composition
- Personnel composition
- Map



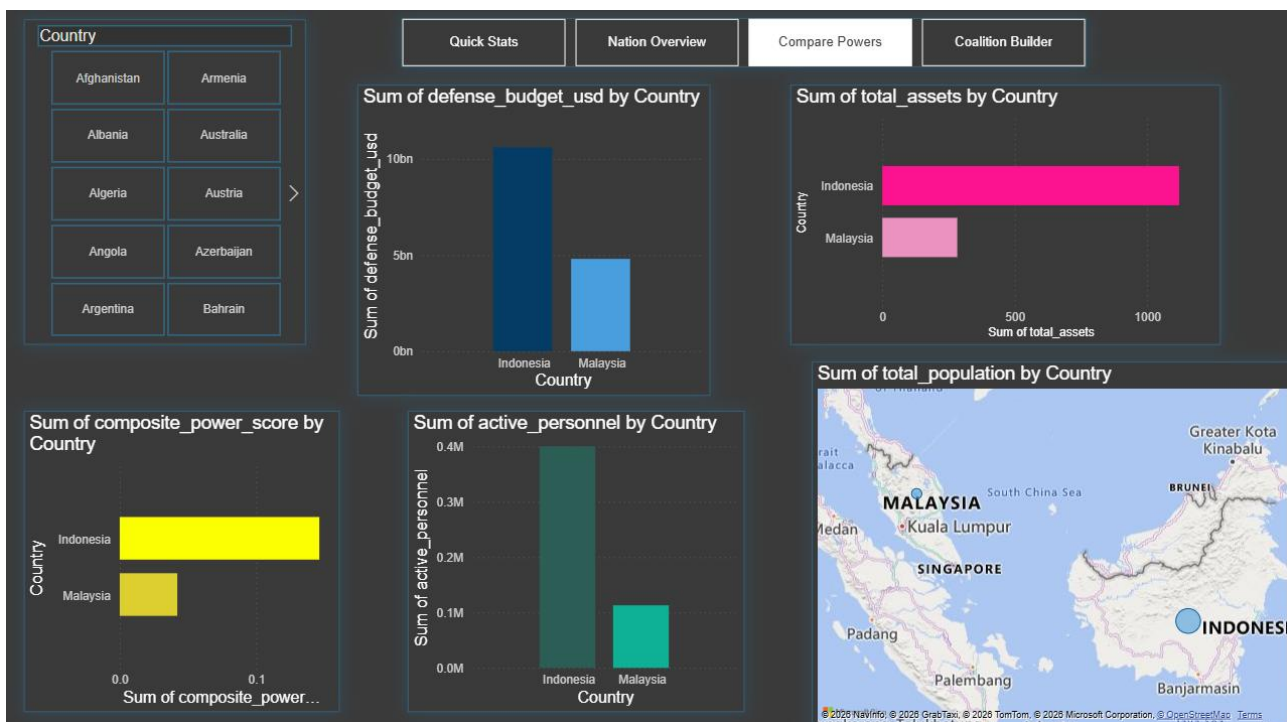
7. Module 6 – Compare Powers and Coalition Builder

7.1 Compare Powers Dashboard

Allows country comparison.

Includes:

- Budget comparison
- Personnel comparison
- Asset comparison
- Power score comparison
- Map comparison

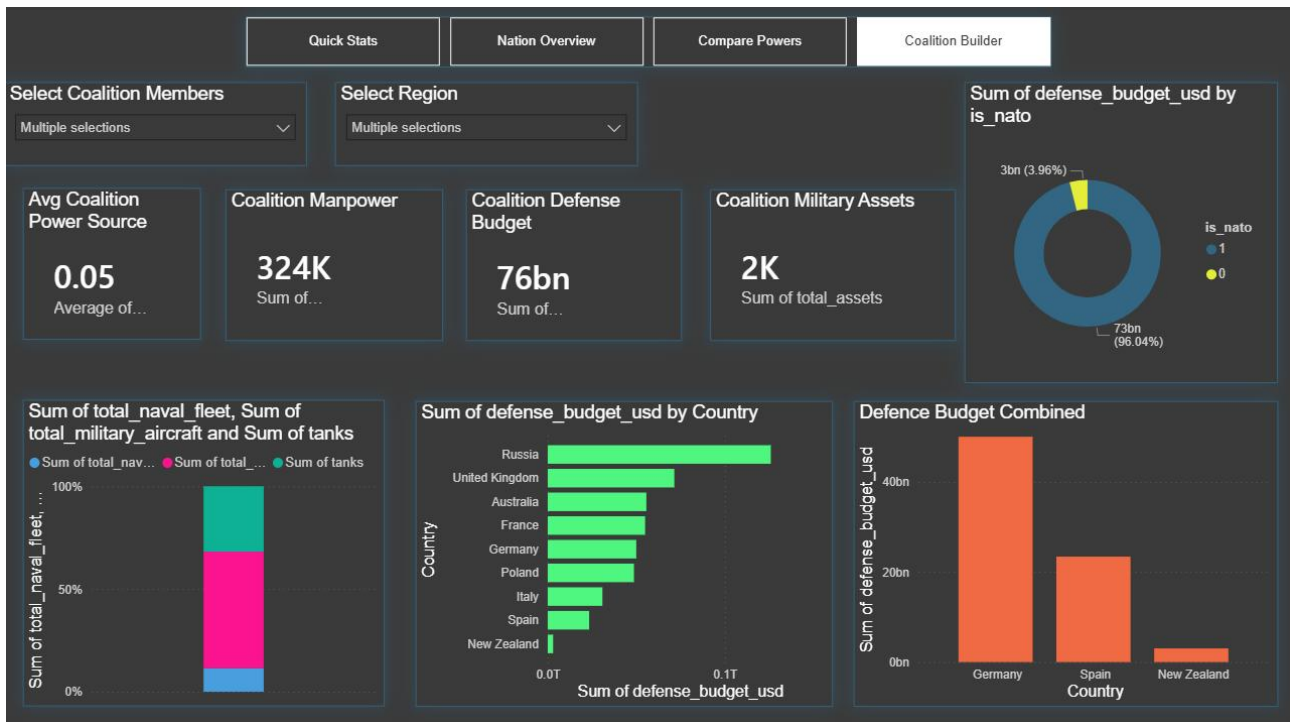


7.2 Coalition Builder Dashboard

Simulates alliances.

Includes:

- Multi-country selection
- Coalition manpower
- Coalition budget
- Coalition assets
- NATO contribution
- Budget contribution
- Asset composition



8. Module 7 – Testing and Debugging

Testing included:

- Filter validation
- KPI validation
- Navigation testing
- Coalition testing

Manual verification ensured correct calculations.

9. Module 8 – Documentation and GitHub Release

Repository includes:

- /scripts
- /data
- /dashboard
- /docs

Contains:

- Python scripts
- datasets
- dashboards
- reports

10. Conclusion

This project demonstrates a complete data analytics pipeline from data collection to visualization. The Unified Military Analytics Dashboard provides a comprehensive platform for military comparison and coalition simulation.

The project demonstrates skills in:

- Web scraping
- Data cleaning
- Feature engineering
- Dashboard design
- Data visualization

The final system provides an interactive environment for analyzing global military power in 2025.